Rebecca Wilson

Applied Research Engineer | Amsterdam, NL
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Summary

Hands-on engineer who designs, 3D prints and tests hardware, then takes it into the field. At Laser Precision Solutions I design mounts and fixtures in CAD, print them myself, build and instrument test rigs, and analyse test data in Python. Before that I commissioned and maintained laser cleaning systems on four US railways. Engineering graduate (University of Glasgow) with coursework in aerodynamics, aircraft design, propulsion and control systems. English (native) and Dutch (B1).

Key Skills

- **CAD & 3D Printing:** Fusion 360, SolidWorks, functional 3D-printed parts (mounts, fixtures, brackets), machining (lathe and mill), soldering and wiring
- **Testing & Field Work:** Test rig and fixture design, sensor integration, field commissioning and troubleshooting, live-track trials
- **Software & Data:** Python (pandas, OpenCV) for data processing and test scripts, Raspberry Pi and Linux, MATLAB
- **Communication:** Test procedures, technical reports, presentations to customers and engineers

Experience

Applied Research Engineer - Laser Precision Solutions (LPS) | Aug 2026 - Present

- Technical lead for laser ablation testing: design mounts, fixtures and brackets in CAD and 3D print them, source components, and build and instrument test rigs with optical and laser-based sensors.
- Standardised a laser cleaning test method that reliably removes 25 µm of contamination in a single pass, now extending testing toward 50 µm.
- Technical lead for a 4-day live-track field trial with Network Rail (UK) in December 2026, leading a team of 2-3.
- Leading concept design of a compact compressor and chiller package for a tram deployment, targeting a 30-50% smaller footprint than the train-based setup.
- Automate results processing and out-of-spec flagging in Python, write formal test procedures and client reports, and guide a data analyst day to day.

Application Engineer - Laser Precision Solutions (LPS) | Apr 2025 - Aug 2026

- Supported LaserTrain deployments for the Massachusetts Bay Transportation Authority (MBTA), Metro-North Railroad, Long Island Rail Road and Staten Island Railway, from installation and commissioning to testing and handover.
- Key technical contact between customers and internal engineering, product and software teams, coordinating multiple projects across distributed teams.
- Supported customer visits, demonstrations and industry conferences, including InnoTrans and RailTech Europe 2026, and hosted a Young Rail Professionals visit.

Selected Projects

Automated Camera and Image Analysis System - ring test rig, LPS

- Built a Raspberry Pi camera system that detects 16 ArUco markers on a rotating ring and captures an image automatically when each marker is centred and still; designed and 3D printed the camera mount through three versions to fit the rig and keep ports accessible.
- Wrote Python (OpenCV) analysis that corrects for lighting drift against a reference card and measures contamination coverage against each marker's clean baseline, combined with gauge thickness readings, to support fluence testing across the laser cleaning band.

Laser Cleaning Validation - European rail operator

- Designed a test method using a rotating test rig and rail tribometer to measure friction before and after cleaning at average and worst-case contamination thicknesses; confirmed removal at real-world speeds with friction restored to near clean rail.

Education

BSc Aeornautical Engineering | University of Glasgow | Sep 2019 - Jun 2024

- Key areas: propulsion, aircraft design, control systems, aerodynamics.